



Shark Attacks and the Poisson Approximation

by Byron Schmuland[†]



A story with the dramatic title “*Shark attacks attributed to random Poisson burst*” appeared on September 7, 2001 in the National Post. In the story, Professor David Kelton of Penn State University used a statistical model to try to explain the surprising number of shark attacks that occurred in Florida last summer. According to the article, Kelton suggests the spate of attacks may have had nothing to do with changing currents, dwindling food supplies, the recent rise in shark-feeding tourist operations, or any other external cause.

“Just because you see events happening in a rash like this does not imply that there’s some physical driver causing them to happen. It is characteristic of random processes that they exhibit this bursty behaviour,” he said.

What was the professor trying to say? Can mathematics really explain the increase in shark attacks? And what are the mysterious Poisson bursts?

The main point of the Professor Kelton’s comments was that unpredictable events, like shark attacks, do not occur at regular intervals as in Figure 1(a), but tend to occur in clusters, as in Figure 1(b). The unpredictable nature of these events means that there are bound to be periods with a higher than average number of events, as well as periods with a lower than average number of events, or even no events at all.

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http://www.stat.ualberta.ca/people/schmu/schmu/dept_page.html

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Figure 1

The statistical model used to study the sequences of random events gets its name from French mathematician Siméon Denis Poisson (1781-1840), who first wrote about the *Poisson distribution* in a book on law. The Poisson distribution can be used to calculate the chance that a particular time period will exhibit an abnormally large number of events (Poisson burst), or that it will exhibit no events at all. Since Poisson’s time, this distribution has been applied to many different kinds of problems, such as the decay of radioactive particles, ecological studies on wildlife populations, traffic flow on the Internet, etc. Here is the marvellous formula that helps to predict the probability of random events:

The chance of exactly k events occurring is

$$\frac{\lambda^k}{k!} \times e^{-\lambda},$$

for $k = 0, 1, 2, \dots$

The funny looking symbol λ is the Greek letter lambda, and it means “the average number of events.” The symbol $k! = k \times (k-1) \times \dots \times 2 \times 1$ is the factorial of k , and $e^{-\lambda}$ is the exponential function e^x with the value $x = -\lambda$ plugged in. Let’s take this new formula out for a spin.

Shark Attack!

If, for example, we average two shark attacks per summer, then the chance of having six shark attacks next summer is obtained by plugging $\lambda = 2$ and $k = 6$ into the formula above. This gives

$$\text{probability of six attacks} \approx (2^6/6!) \times e^{-2} = 0.01203,$$

which is a little more than a 1% chance. This means that six shark attacks are quite unlikely in any one year, although it is likely to happen about once every 85 years. The chance that the whole summer passes without any shark attacks can also be calculated by plugging $\lambda = 2$ and $k = 0$ into the formula. This gives

$$\text{probability of no attacks} \approx (2^0/0!) \times e^{-2} = 0.13533,$$

which is a 13% chance. Thus, we can expect a “sharkless summer” every seven or eight years.

In this hypothetical shark problem, the number of attacks followed the Poisson distribution exactly. The Poisson distribution is most often used to find approximate probabilities in problems with n repeated trials and probability p of success. Let me show you what I mean.